CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents
Supplementary Material

Anonymous submission

Scope of the Supplement

The main paper is self-contained. This supplementary document provides the complete fixed-budget design, confidence intervals for the policy and selector ablations, per-round closed-loop results, and deployment-cost measurements. The supplement is a separate PDF and does not count toward the seven pages of main content. All results use the same summary trace and archived screenshots as the main paper. Restoring an event means retrieving its archived post-action screenshot and reattaching it to the event summary; no screenshot is generated from text.

Fixed-Budget Restoration Protocol

At decision t, every completed event remains visible as an action summary. An active visual-context budget B allows exactly B summaries to be promoted to summary-plus-image form. Recent-B promotes the latest B eligible events. A restoration allocation may instead demote a recent event to summary-only form and promote a distant event, preserving the image count, image resolution, summary trace, and prompt structure. The realized number of non-recent promotions is

$$k = |S \setminus \text{Recent-}B|.$$

It is an observed selector output, not a preset experimental treatment.

B	Recent arm	Replacement training arm	Deployed allocation
1	Recent-1	recent → archived	Selected-1
2	Recent-2	oldest → archived	Selected-≤ 2
4	Recent-4	oldest → archived	Selected-≤ 4
8	Recent-8	built, not trained	Selected-≤ 8

Table S1: Matched-budget fidelity-restoration design. The B=8 row is a budget-extrapolation test; it is excluded from adapter and selector training.

Intervention arms. For a matched group, R_0 and R_A use Recent-B under the frozen and adapted policies; S_0 and S_A replace the oldest recent image with a target-relevant archived image; W_A uses an age-matched wrong archived image. The adapter selection statistic is

$$\text{DiD}_{\text{select}} = (S_A - S_0) - (R_A - R_0).$$

The gate requires a positive episode-clustered bootstrap lower bound and point-estimate drift caps $|R_A - R_0| < 0.02$ and $|W_A - W_0| < 0.02$. Reported coalition utilities always rerun the complete restored set; they are never sums of singleton utilities.

Uncertainty. Offline intervals use 2,000 episode-clustered bootstrap replicates. Closed-loop intervals resample tasks while preserving the paired arms and repeated-run structure. A confidence interval crossing zero is described as indistinguishable, not as evidence for equivalence.

Additional Desktop Results

Policy-Side Gates Across Visual Budgets

Adapter	B	$\text{DiD}_{\text{select}}$ [95% CI]	$\lvert A_r\rvert$	Wrong	Gate
HGKV	1	$+.0224$ $[+.0162, +.0294]$	.0088	.0085	pass
HGKV	2	$+.0164$ $[+.0110, +.0221]$	.0107	.0078	pass
HGKV	4	$+.0109$ $[+.0070, +.0154]$	.0098	.0091	pass
HGKV	8	$+.0063$ $[+.0018, +.0115]$	.0106	.0137	pass
Ungated	1	$+.0177$ $[+.0135, +.0220]$	.0086	.0071	pass
Ungated	2	$+.0139$ $[+.0102, +.0177]$	.0079	.0069	pass
Ungated	4	$+.0075$ $[+.0031, +.0118]$	.0098	.0089	pass
Ungated	8	$+.0047$ $[+.0007, +.0093]$	.0160	.0163	pass

Table S2: Complete per-budget adapter gates. Group counts for $B = 1/2/4/8$ are $94/72/45/11$; the frozen selection effects are $+.0791/+.0256/-.0002/-.0305$. B=8 is untrained.

Both adapters pass, but HGKV has a larger selection effect at every budget and remains farther from the drift caps at B=8.

Why the recent reference matters. At B=1, the deployment renderer's most recent slot can duplicate the current screenshot. Against that redundant control, the frozen policy prefers the relevant archived event by $+0.0333$ nats/token (95% CI $[+0.0118, +0.0567]$). Against the informative true previous frame, the frozen preference is only $+0.0013$ $[-0.0242, +0.0253]$. The HGKV adapter retains a positive

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true-previous-frame DiD of $+0.0095$ $[+0.0056, +0.0138]$. Thus the learned selectivity is not an artifact of comparing useful pixels with a duplicated current screen.

Selector and Feature Ablations

Allocation rule	B=2	B=4
Learned scorer	$+.0297$ $[+.0185, +.0409]$	$+.0169$ $[+.0067, +.0277]$
Random exact-B	$+.0058$ $[-.0056, +.0168]$	$+.0082$ $[-.0009, +.0177]$
RGB similarity	$+.0061$ $[-.0032, +.0150]$	$+.0071$ $[-.0010, +.0152]$
Inverted score	$-.0159$ $[-.0290, -.0033]$	$-.0091$ $[-.0180, -.0004]$
No recency features	$+.0121$ $[-.0003, +.0246]$	$+.0084$ $[-.0012, +.0173]$

Table S3: Exact-B selector ablations with fresh full-set policy scoring. Random and RGB-similarity rules do not explain the gain; reversing the learned ranking is harmful.

At B=4, the learned scorer chooses $k = 0/1/2/3/4$ on $103/103/88/119/146$ evaluated states. It therefore retains the entire recent window on 19% of states and performs non-recent restoration selectively. The corresponding distributions for random allocation are $96/138/161/152/48$, and for RGB similarity $233/166/106/66/24$. Performance cannot be attributed to using a fixed sparse history pattern or to always displacing recent images.

Witness-reference ladder. The single-pass last-action reference yields $+0.0247$ $[+0.0133, +0.0363]$ at B=2 and $+0.0152$ $[+0.0058, +0.0250]$ at B=4 on desktop. Removing the witness reduces these effects to $+0.0161$ $[+0.0025, +0.0299]$ and $+0.0086$ $[-0.0024, +0.0196]$, respectively. This motivates the proposal-conditioned two-pass path as the default for cross-platform transfer, while retaining last-action conditioning as a domain-matched efficiency variant.

Additional Closed-Loop Results

Zero-Shot Mobile Evaluation

Method	Run 1	Run 2	Run 3	Mean
Frozen, summary-only	$33/117$	$32/117$	$34/117$	28.2%
HGKV + Recent-4	$34/117$	$37/117$	$35/117$	30.2%
CausalCache-P	$39/117$	$42/117$	$38/117$	33.9%

Table S4: Per-round zero-shot MobileWorld success. The paired CausalCache-P minus HGKV+Recent-4 mean is +3.7pp (95% CI $[-0.9, +8.6]$, two-sided $p = 0.129$).

Stratum	CausalCache-P	Recent-4	Δ
Memory-critical	28.0%	19.4%	$+8.6$pp $[+1.6, +16.1]$
Control	40.6%	42.4%	-1.8pp $[-6.7, +3.6]$

Table S5: Primary mobile allocation analysis ($n = 62$ memory-critical; $n = 55$ control). The split-by-method interaction is $+10.4$pp (95% CI $[+1.5, +19.6]$, two-sided $p = 0.022$).

Outcome-defined sensitivity analysis. On the 95 tasks where the frozen summary-only policy reaches the official step cap, CausalCache-P scores 32.6%, Recent-4 28.9%, and summary-only 23.9%. The paired CausalCache-P minus summary-only difference is $+8.8$pp $[+1.8, +16.0]$; the paired difference from Recent-4 is $+3.7$pp $[-2.6, +10.0]$. On the 22 tasks where summary-only terminates early, CausalCache-P is 43.2%, Recent-4 is 36.4%, and summary-only is 47.0%; CausalCache-P minus summary-only is -3.8pp $[-20.5, +12.1]$. These outcome-defined strata are secondary because they condition on baseline behavior; the construction-defined split in Table S5 is the primary analysis.

Two-pass mobile cost. Across 2,224 audited steps, the proposal-conditioned second policy pass fires on 81.7% of all steps and 96.9% of steps with more than B eligible events. Median pass-1, selector, and conditional pass-2 times are 4.229 s, 0.039 s, and 3.258 s, respectively. Mean per-step overhead relative to pass 1 is 65.9%. The selector itself is cheap; the extra policy pass dominates.

OSWorld-Verified Evaluation

Method	Rate	Paired comparison
B0	19.9% (72)	reference
FR	32.7% (118)	$+12.7$pp $[+8.6, +17.2]$ vs. B0
HR	33.0% (119)	$+0.3$pp $[-3.1, +3.6]$ vs. FR
CausalCache-LA	34.1% (123)	$+1.4$pp $[-2.5, +5.0]$ vs. FR
CausalCache-P	32.7% (118)	-0.3pp $[-3.9, +3.3]$ vs. HR

Table S6: OSWorld-Verified closed-loop results on the official no-GDrive 361-task roster with a 15-step budget; parenthesized values are success counts. High-fidelity history is beneficial relative to summary-only memory; same-budget allocations are statistically indistinguishable. B0 denotes frozen summary-only, FR frozen Recent-4, and HR HGKV Recent-4.

For CausalCache-P on OSWorld, the median task incurs eight extra policy calls and 20.6 model-seconds. Median wall time is 307.6 s, so the added model time is approximately 7% of end-to-end task time. These results separate two claims: desktop closed loop supports the availability of high-fidelity history, while the mobile construction-defined split tests which summarized events should receive that fidelity.